



Deliverable D3.8. A report on dispersal-mediated biodiversity effects on ecosystem functioning in intermittent rivers

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Abstract

Different aquatic groups can regulate multiple ecosystem functions. However, the effects on ecosystem functions by interactions between different taxonomic groups operating on various functional levels are poorly known, especially in drying, intermittent rivers. Stream drying, as one of the most important stressors today, shapes communities inhabiting these ecosystems and substantially influences many community characteristics during and after drying, e.g. diversity, abundance, biomass, and productivity. Direct and indirect effects of drying-induced structural and functional reduction are subsequently remediated during the recolonization following flow resumption. This recolonization happens by various pathways (upstream or downstream migration, hyporheic or aerial recolonization), with potential implications for stream ecosystem functions through top-down or bottom-up interactions among key components differing in dispersal ability or dispersal pathways (e.g. macroinvertebrates and biofilms including algae, bacteria and fungi). As little is known about the importance of particular recolonization pathways, we designed an experiment focused on the description of the contribution of different recolonization pathways to post-drought remediation of stream functions. A great effort has been made to improve the experimental design, especially in construction and tuning the design of in-situ experimental devices that allow only one recolonization pathway at a time. The design and functionality of these devices were proven during the pilot experiment and subsequently adjusted based on our findings.

Only weeks after its start, the core experiment was unfortunately destroyed by an unpredictable flash flood with an approximate 50-year recurrence probability, so only partial results from the pilot

experiment in early spring can be presented in this deliverable. During the pilot experiment, we examined the functionality of the proposed experimental units and the contribution of the four mentioned recolonization pathways to the establishment of macroinvertebrate communities in two weeks. Our results show that the most important pathways for fast reestablishment of the community are downstream drift and vertical migration from hyporheic refuges. The latter plays a dominant role in relation to biomass. The observed minor contribution of aerial colonisation could originate from the insufficient duration of the experiment and the short time period available for the hatching of eggs laid by flying adults. The pilot experiment shows the importance of several advantageous species traits enabling different colonization ways, such as a typical occurrence in drift for downstream colonization or affinity to coarse substrates for hyporheic colonization of experimental units. A life cycle with an aerial stage is likely also a relevant trait.

The pilot study results proved the suitability of the applied technical design of the experiment. The preliminary results suggest effective mechanisms of community recovery in streams affected potentially by drying according to known climate change scenarios, thus our results from the DRYvER pilot study proved that this experimental design could be replicated in future studies to explore further effects of different recolonization pathways on the resilience of fluvial ecosystems to drying.

Keywords

Biodiversity, ecosystem function, recolonization, dispersal, drying river networks, invertebrates,

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¹ Use one of the following codes:

R=Document, report (excluding the periodic and final reports)
 DEM=Demonstrator, pilot, prototype, plan designs
 DEC=Websites, patents filing, press & media actions, videos, etc.
 OTHER=Software, technical diagram, etc.
 ORDP : Open Research Data Pilot

² Use one of the following codes:

PU=Public, fully open, e.g. web
 CO=Confidential, restricted under conditions set out in Model Grant Agreement
 CI=Classified, information as referred to in Commission Decision 2001/844/EC.

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Introduction

Metacommunity concept

A *metacommunity* is characterised as a set of local communities connected by species dispersal. Beyond connectivity among local communities, a metacommunity's structure is likely to depend on the present biotic groups with specific traits related to the environmental characteristics of the habitats (Leibold et al. 2004; Heino et al. 2015a). The origins of metacommunity theory accentuated distributed communities in space connected via dispersal. Metacommunity ecology aims to understand the shaping of local communities in a regional context (Heino et al. 2015b), i.e. through connection of local communities via the dispersal of several potentially interacting species. The current theory usually considers four perspectives: species sorting, mass effects (or isolation), patch dynamics and neutral model. The species sorting emphasises links between the environmental templates and selected species traits. When such species filtering based on local environmental factors dominates, species typically occur in environmentally suitable habitats. In contrast, mass effects (or isolation) reflect the dispersal-mediated persistence (or non-occurrence) of species without considering such a match of species traits with habitat features. The patch dynamics concept stresses competition-colonisation trade-offs, i.e. it downplays species-environment interactions yet does not regard species as neutral. This contrasts the neutral model, which finally suggests traits to be irrelevant to community structure, and metacommunity substructuring to thus purely emerge from random extinction and recolonization events (Perkin et al. 2021).

Advanced metacommunity research indicates that patch dynamics and mass effects are subcases of species sorting with limiting and homogenising effects of dispersal (Winegardner et al. 2012). Thus, the focus in metacommunity studies should be stressed on the relative roles of *species sorting* and *dispersal*, representing fundamental processes in all metacommunity structuring (Lindstrom & Langenheder 2012). Yet, the level of interaction between dispersal limitation and such environmental regulation differs among various ecosystems and spatiotemporal scales (Heino et al. 2015a). Among aquatic “meta-systems” such as river networks, one can observe differences in connectivity and environmental heterogeneity, suggesting that metacommunity organisation might differ among thereby contrasting systems.

Dispersal in drying river networks as spatiotemporal barriers

The dispersal abilities of aquatic organisms and their travel distances are generally underexplored in major biotic groups. This deficit is a principal obstacle to understanding dispersal as a regional process structuring ecological communities (Havel & Shurin 2004). Freshwater invertebrates are commonly widely spread inhabitants of inland aquatic systems able to reflect changes in environmental conditions with fast response at the species, population or community level (Stubbington et al. 2019a, 2021). Freshwater invertebrates have been characterised as frequent, widespread dispersers, particularly those transported passively. However, this characterisation may stand only for dispersal *potential*, not for generally manifested *actual* dispersal rates that may be highly variable among environmental habitats and taxonomic groups (Bohonak & Jenkins 2003). The persistence in repeatedly disturbed environments, as are drying river networks (DRNs), could be achieved through *resistance* and *resilience* strategies, which promote community recovery to a pre-disturbed state (Datry et al. 2017).

Flow intermittence represents a hydrological disturbance acting as an environmental filter and intense selection pressure on instream biota (Lytle & Poff 2004). Drying is a crucial driver of invertebrate population dynamics and can result in direct loss of species richness and shifts in species composition (Bogan & Lytle 2011), as well as altering key ecosystem functions such as leaf litter decomposition (Corti et al. 2011).

Drying produces a dual response of freshwater invertebrates at various spatial scales. On a local, ecosystem scale, it reduces the number of drying-sensitive taxa (Arscott et al. 2010) and, therefore, lessens alpha diversity within a drying stretch. At the regional scale of river networks, drying typically generates a nestedness pattern in spatiotemporal beta diversity, where communities of non-perennial reaches become subsets of those at perennial sites (Datry et al. 2014; Stubbington et al. 2019b). Regarding temporal variation, drying may provoke taxa turnover, often by complete species replacement, in non-perennial reaches subjected to strong environmental variation due to the succession of dry and wet phases. Applying the metacommunity concept on DRNs and their biota might provide the capability to predict potential metacommunity shifts as ecological responses to future alterations of flow regime associated with on-going climate change. Components of beta diversity (i.e., nestedness or turnover) enable to presume metacommunity structuring mechanisms, where i) primarily nested metacommunities derive from dispersal limitation while ii) metacommunities with higher turnover develop from environmental filtering. The structuring mechanisms assume nested metacommunities to be most prevalent during wet periods, while species turnover patterns should dominate during the drying, and rewetting phases (Datry et al. 2016). In addition, this taxonomic spatiotemporal habitat sharing promotes an overall gamma (regional) diversity in river basins, or even further in the surrounding landscape, including terrestrial species (Corti & Datry 2016).

Dispersal and recolonization strategies in drying river networks

The recolonization strategies of DRN inhabitants are enhanced with specific species traits, reflecting biological and ecological adaptations to environmental conditions and characterising functional community composition. Resistance and resilience are vital strategies of organisms to cope with environmental stressors, which allow species, populations, and communities to persist long-term in repeatedly disturbed ecosystems (Lake 2013).

In the context of DRNs, the resistance strategy represents the ability to withstand a drying period at the place of its impact (Chester & Robson 2011), typically by using a refuge. Besides various moist or wetted mesohabitats and residual pools within a drying reach, the most effective instream refuge appears to be the hyporheic zone (Dole-Olivier 2011). The hyporheic zone represents water-saturated sediments and interstitial pores beneath the streambed with lateral extension into banks and floodplains (Stubbington 2012). This zone is a dynamic ecotone between surface and groundwater systems and plays a crucial role in the functioning of lotic ecosystems. It forms a critical refuge in gravel-bed streams, where large interstitial spaces among substrate grains enable active or passive vertical migration to watered parts during surface drying (Vadher et al. 2017; Loskotová et al. 2019). The hyporheic zone thus provides an important *in situ* source for recolonization after rewetting, when resilience strategies come into play additionally.

The resilience mechanism is particularly understood as the ability of organisms to rapidly recolonize a disturbed site after the end of a drying period (Hershkovitz & Gasith 2013). Further, resilience can relate to disturbance avoidance by leaving the drying site in advance. Many invertebrate taxa avoid the gradual drying by various mechanisms, such as i) active or passive drifting downstream to perennial reaches (Dewson et al. 2007) or ii) crawling and swimming upstream or downstream to such refuges (Lytle et al. 2008). Moreover, some insect taxa have developed an ability to iii) emerge as aerial adults before drying and consequent oviposition or adults return after the drying is over (Bogan & Boersma 2012). Together with *in situ* refuges (i.e., hyporheic zone, remnant pools), all these mechanisms constitute sources of recolonizers during the rewetting phase (Chester et al. 2015).

Apparently, poor dispersers must endure the presence of a disturbance and re-establish population from *in situ* sources, while organisms with good dispersal abilities are likely resilient. The resilience of invertebrate populations following flow resumption can be high, reducing the long-term effects of river drying on biodiversity and ecosystem functioning. Full recovery of an invertebrate community to its pre-disturbed state may vary, but generally occurs within a few weeks or several months (Churchel & Batzer 2006; Pařil et al. 2019). Nonetheless, the recolonization pathway preference of each taxon is

still unknown and almost impossible to detect through simple field observation (Vander Vorste et al. 2016).

Finally, the implications of drying on the recovery of macroinvertebrate communities can have cascading effects on in-stream ecosystem functioning. Drying may strongly alter ecosystem processes through its effects on the biomass, diversity and functional trait composition of local biotic communities. The resumption of ecosystem functioning after drying is likely to be impeded by the dispersal limitations acting on biota, which will depend on the dispersal capacities of the constituent organisms associated with the regional species pool at the river network scale. Thus, an experimental study designed to understand the effects of post-drying recolonization through particular dispersal mechanisms of freshwater invertebrates in DRNs and their later implications for ecosystem functioning was the main aim of this deliverable.

Objectives and Hypotheses

The regional macroinvertebrate species pool may differ in terms of adaptation to drying across Europe with respect to the long-term presence or absence of drying, i.e., novel vs. historical drying networks sensu (Crabot et al. 2021). We expected a harsher impact of dry episodes on macroinvertebrate communities and associated ecosystem functions in more novel DRNs. Following this logic, we selected as study sites two reaches within the DRN of the *Velička* river in the continental humid climate of Central Europe, where drying streams have emerged recently in the last two decades (Pařil et al. 2019; Straka et al. 2019). Within the network, the two study reaches were carefully selected to differ strongly in their flow regimes (i.e., perennial vs non-perennial), and thus on their species pool available for recolonization. This way, we aimed to promote different recolonization strategies of macroinvertebrates after a simulated drying scenario based on the capability of the species pool to resist or recover from drying stress.

Main research questions and hypotheses

We hypothesized that spatial constraints in dispersal associated with drying will cause a reduction in multi-trophic diversity (specifically invertebrate community composition), with implications for ecosystem functioning through a decrease in i) the rates and magnitude of biogeochemical processes and ii) the resilience of these processes. The experiment was designed to test the following hypotheses:

H1: The pathway of invertebrate recolonization affects multi-trophic diversity and functional recovery of stream communities following drying.

H2: The flow regime shapes the species pool of colonizers and, therefore, the relative importance of recolonization pathways for multi-trophic diversity and functional recovery of stream communities following drying.

Following our rationale, we derived the following questions:

Q1: How does the resilience of functions after drying depend on the recolonization of functional groups with different dispersal pathways?

Q2: How does that depend on the flow regime of the upstream river network?

In the proposed experiment, we focused on the indirect, dispersal-mediated effects of drying via its impact on macroinvertebrate abundance and diversity and how this, in turn, affects the ecosystem functioning as a result of different community compositions.

The results should lead to the evaluation of the roles of different functional groups with contrasting dispersal modes through experimental manipulation of recolonization through selective blocking of dispersal pathways. Functional effects will mostly be top-down effects (e.g., on periphyton biomass). The experiment also provides an opportunity to assess the effects of diversity on function at the ecosystem level. Diversity consequences may be direct effects of colonization (when assessing the diversity of invertebrates) or indirect top-down effects when assessing the diversity of bacteria or algae.

Methods

Experimental design and phase schedule

The experiment was designed to make use of *in situ* exposed experimental flumes with manipulated, i.e., selectively allowed/blocked, dispersal pathways. We recognized three fundamental consecutive phases in such an experiment:

T0: Colonization phase during a flowing phase without any dispersal control, with a minimum duration of 2 weeks, to win an overview over the locally available species pool (see below). The aim is to measure the average and variance of the community structure and functions and demonstrate similarity across replicates despite potential local influences or stochastic effects in the relatively small flumes. First sampling period.

T1: Artificial drying phase at least 1 week to eliminate all non-resistant invertebrates. We expect microbial communities to have higher survival chances in such experimental drying.

T2: Rewetting phase with dispersal controls installed (see below) divided into two subphases:

T2a: Early 1-week recolonization after re-establishment of flowing conditions. Microbial communities may have regained function to some extent, but invertebrate recolonization may be limited to faster dispersers and more efficient dispersal pathways. Second sampling period.

T2b: Long-term recolonization expected in 3-4 weeks after re-establishment of flowing conditions. By this time, dispersal treatments should have had a clear effect on invertebrate recolonization and potentially further effects on the microbial community. Third sampling period.

We considered four possible recolonization pathways (i.e., downstream, upstream, hyporheic and aerial). Each experimental treatment thus allowed only one pathway while all other pathways were blocked. A control group with no path blocked was added as a separate treatment. Taken together, the experiment used five treatments:

D: only **downstream** waterborne dispersal mostly by passive drift: nets along all sides and top, except at the upstream side

U: only **upstream** waterborne dispersal mostly by active crawling: nets along all sides and top, except at the downstream side

H: only upward **hyporheic** dispersal: nets along all sides and top except on the bottom

A: only **aerial** dispersal: nets along all sides but no nets at the top

C: no dispersal constraints: no use of nets as barriers, **control** treatment.

All the treatments were provided with four repetitions for each of two main sampling periods: i) the early recolonization after drying (**T2a**; 20 units) and ii) the fully recolonized/recovered community (**T2b**; 20 units). Four control units were planned to sample before the drying simulation (**T0**). This experimental design was applied in parallel in the perennial and non-perennial stream reaches. Taken together, we installed 88 experimental units.

Experimental equipment – box units

The experimental units were designed as paired sets of outer and inner wired stainless-steel boxes (mesh size 25 x 25 mm, 3 mm wire thickness for inner box and 50 x 100 mm mesh size with 4 mm wire thickness for outer box), with top openings and the following dimensions (WxDxH): outer box 360 x 250 x 200 mm; inner box 310 x 240 x 120 mm. All outer boxes were covered from both sides with PVC panels to prevent side entry of the invertebrates, so that only upstream/downstream/aerial/hyporheic paths were open in a fully opened box. Wired boxes were designed to provide stability in flowing water and to avoid debris, logs and leaf inputs.

All the dispersal constraints were realised using 250 µm netting placed on top, upstream and downstream ends of the units or towards the bottom part (i.e. hyporheic obstacle). The usage of nets implies having to deal with a decrease in flow velocity and flow through the box caused by fine netting and additional blocking of nets by clogging. Therefore, front apertures for adjustments of comparable discharge through boxes were installed to avoid treatment-induced bias of flow through the boxes. Based on previous measurements of discharge with different aperture sizes and types, we selected a compact PVC bar at 60% water level depth on the front side of the outer box as an optimal aperture limiting water entry.

Study sites

The actual *in situ* experiment was planned to be conducted in the Czech Republic at two hydrologically different tributaries of the main stem of the *Velička* river within the model DRN catchment (White Carpathians Landscape Protected Area). The first tributary, *Hrubý potok*, was selected as a fully perennial stream network with no observed non-perennial reaches in the catchment. The second tributary, *Kazivec*, was selected as a non-perennial stream with a high proportion of days without surface flow (flow intermittence 10 % of the year) in the catchment (Fig 1). Both catchments have comparable size of catchment, hydromorphology and geology. The experiment was planned to be done in parallel in both streams according to the natural occurrence of dry conditions within the season for the flowing-drying-rewetting phases. The experiment was successfully started in the middle of April 2024, with T0 completed and T1 set up. However, just before the T1 phase termination, an unexpected flash-flood occurred, which completely destroyed the *in situ* experimental units and prohibited the continuation of the experiment (see Results for details).

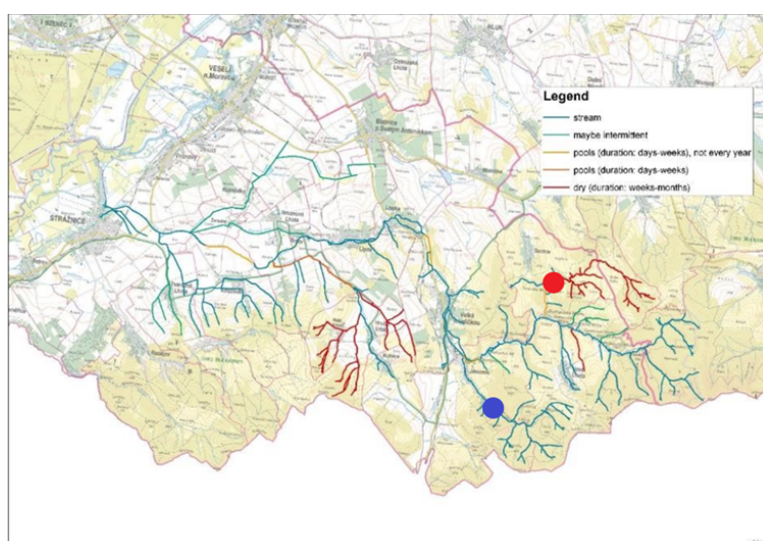


Fig 1. Map of Velička DRN catchment with *Hrubý potok*, perennial experimental site (blue circle) and *Kazivec*, non-perennial experimental site (red circle).

Pilot study

To properly adjust such a complex experiment, we created a simplified design for a shorter pilot study, which also served as a trial to test the experimental design itself and material functioning in real in-stream conditions. We added to the previously described five units with flow/current regulation a differently shaped second control unit (coded as CA) which uses front apertures for better control of discharge through the unit (see further). In the pilot study, 18 custom-made experimental boxes of six different dispersal treatments in three replicates were used as experimental units, which were created to artificially enable only one of the supposed recolonization pathways. Together, we obtained here six treatments:

A: only **aerial** dispersal,

H: only upward **hyporheic** dispersal,

D: only **downstream** waterborne dispersal,

U: only **upstream** active dispersal,

C: no dispersal constraints: **control** without use of nets as barriers

CA: no dispersal constraints: control without nets as barriers, with front **aperture** regulating flow added

The main aims of the pilot study were to

- i) determine colonization times to get a reasonably stable community which may vary among the treatments according to the colonization pathway and
- ii) identify variance in flow rates affected by clogging and find out if using an aperture in the control group is helpful at all to bring flow through the control treatment closer to the dispersal-mediated treatments, which all suffer to some extent from reduced flow under real field conditions.

Furthermore, the pilot study aimed to solve some operational questions such as fixing aperture bars, clogging times and cleaning effort and frequency, handling the destructive sampling, and focusing on getting time/effort estimates for the various operations needed to run the experiment.

The pilot experiment was conducted for two weeks in March 2023 at one of the pre-selected sites, the perennial stream *Hrubý potok*, in the DRN *Velička* river catchment. The pilot study simulated the recolonization phase during rewatering (**T2**).

Pilot study set up

The experimental units were installed in a 200 m reach in serial sequence with randomised unit settings for each spatially designed repetition. In all spatial blocks, each treatment was represented by one unit (Fig 2).



Fig 2. Representation of experimental units installed in serial sequence in the stream.

Each outer box with an inserted inner box was buried 12 cm below the stream bottom level and fastened with tent pegs (Fig 3). The water level above streambed surface in all units was comparable around 15cm. Inner boxes were filled with a mixture of prewashed coarse gravel substrate (diameter 16–32 mm and 32–90 mm, respectively), up to three-quarters of its volume. The gravel substrate represents a platform for biotic colonisation (Fig 3).



Fig 3. One experimental unit (outer and inner box) installed in the stream (left) and inner box filled with mixture of gravel substrate with hyporheic colonization barrier formed by a 250 μ m mesh size net (right).

The dispersal barriers for upstream and/or downstream recolonization paths were set using a frame with fine 250µm mesh size metal netting placed between the outer and inner box at the front and/or back side, respectively (Fig 4). During the pilot experiment, the frames were manually cleaned once in two days to remove clogging and ensure stable current in the boxes. To block the hyporheic recolonization path, the inner boxes were placed into 250µm mesh size nylon netting; similarly, for blocking aerial recolonization, the outer boxes were covered with 250µm mesh size nylon netting from above (Fig 4).



Fig 4. Installed front and back frames with 250µm metal netting to block downstream and upstream recolonization paths (left) and installed 250µm nylon netting on inner box to block hyporheic recolonization path and on the top of the outer box to block aerial recolonization, respectively (right).

During the installation, physico-chemical water parameters in the stretch were measured on two subsequent dates to estimate the influence of different colonization barriers in flumes on physical-chemical parameters. After two weeks, all the units were destructively sampled by abstraction of the inner box. The substrate in the inner box was repeatedly decanted and washed through a 250µm nylon net to separate invertebrates and any accumulated organic material from used artificial substrate. The extracted sample, which consisted of mostly organic material, was fixed in 4% formaldehyde and sorted under a stereomicroscope, and the invertebrates were determined at the lowest possible taxonomic level (predominantly species/genus, in lesser proportion to family). Afterwards, the invertebrates were dried at 70°C for 24 hours in an oven and then weighed to directly estimate biomass for each experimental unit.

Pilot study data analysis

The sampled invertebrate communities were analysed using NMDS (log-transformed data, Bray-Curtis dissimilarity) to visualize potential differences in community composition among treatments. The community composition of each treatment was simplistically visualized using barplots expressing the percentual representation of each family and other higher taxonomic groups in the treatments' communities.

Furthermore, individual treatments were compared with the control group in terms of alpha-diversity indices, i.e. Shannon diversity index values, total species richness and total invertebrate abundance using one-factorial ANOVA. Shannon diversity index is defined as: $H = -\sum_{i=1}^R p_i \ln(p_i)$, where p_i is the proportional abundance of species i , estimated with function *diversity* of “vegan” R package. In addition, we analysed differences among dispersal treatments for traits associated with the ability of aquatic invertebrates to recolonize newly inundated habitats: Body flexibility, Life cycle, Dissemination strategy, Occurrence in drift, Microhabitat preference, Hyporheal occurrence, Locomotion type and Stream zonation. The values of the percentage representation of each trait in the community (i.e. % of individuals possessing the trait) were compared using one-factorial ANOVA, with a focus on pre-defined contrasts between the control and all other treatments.

Core experiment

In total, 88 wire boxes with substrate were installed in two streams on April 18th 2023 for initial acclimatization and colonization. After 4 weeks (May 12th), the T0 phase was ended and 8 replicates were destructively sampled, and the remaining 80 boxes were processed as planned for T1 phase (inner boxes were pulled out of the water for one week-long drying – see Fig. 5). Five days after (May 17th), the whole experiment was flushed down by an unexpected flash-flood (see Results for details).



Fig. 5 – Inner boxes pulled out of the water above water level, covered with fine net to prevent flying adults to lay eggs on wet substrate. Outer boxes were filled with heavy stones to ensure stability and solid base for inner boxes laying on top of them.

Results

Pilot study

The measured physical-chemical parameters at the beginning of the experiment were: water temperature 9°C, pH 8.49, conductivity 453 µS/cm, dissolved oxygen 11.07 mg/l and oxygen saturation 101.4 %. We measured dissolved oxygen after 5 days from the experiment’s initial installation in each box with the similar results as at the time of the installation implying no effect of dispersal barriers on the oxygen conditions within the unit.

Analysis of the aquatic invertebrate communities using NMDS showed clear differences in community composition among treatments (see Fig 6a). The two control groups (with and without aperture) showed a large overlap in community composition and were therefore combined into a single "control" type in all subsequent analyses. In terms of taxonomic composition, two groups, Chironomidae and Crustacea, dominated all treatment types. In the HYPORHEIC treatment, more than 90 % of the individuals were Crustacea (represented by species *Gammarus fossarum*), while in the UPSTREAM treatment, approximately 75% of the individuals were representatives of the family Chironomidae (see Fig 6b).

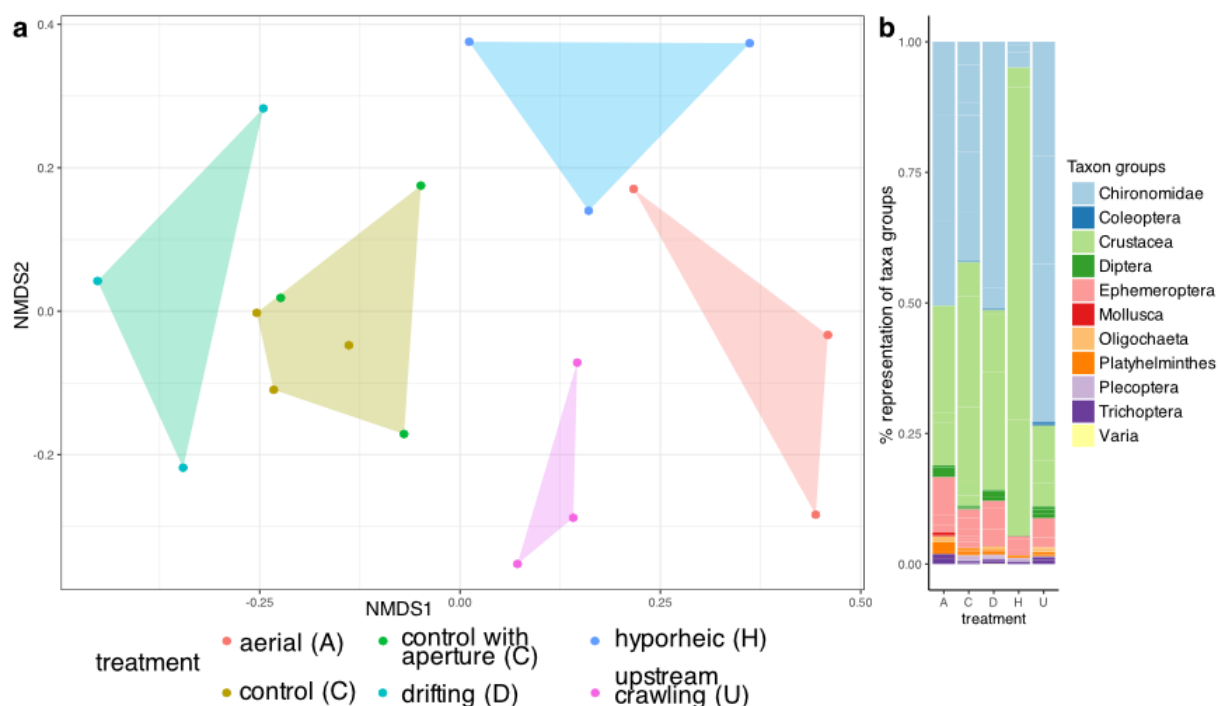


Fig 6. The Non Metric Multidimensional Scaling (NMDS) diagram (a) and barplot of percentage representation of higher taxonomic units (b) across the five different dispersal treatments (control group – C and control group with aperture – CA are considered as one treatment).

Significant differences among treatments were evident in the alpha-diversity indices; Shannon's diversity index (ANOVA; $p = 0.05$), total abundance (ANOVA; $p = 0.007$) and taxa richness (ANOVA; $p = 0.003$). Shannon's diversity index was lowest in the HYPORHEIC treatment, the taxa richness was lowest in the AERIAL treatment, and total abundance was also lowest in the AERIAL treatment (see Fig 7). In other cases, the difference between the control group and any treatment was not significant.

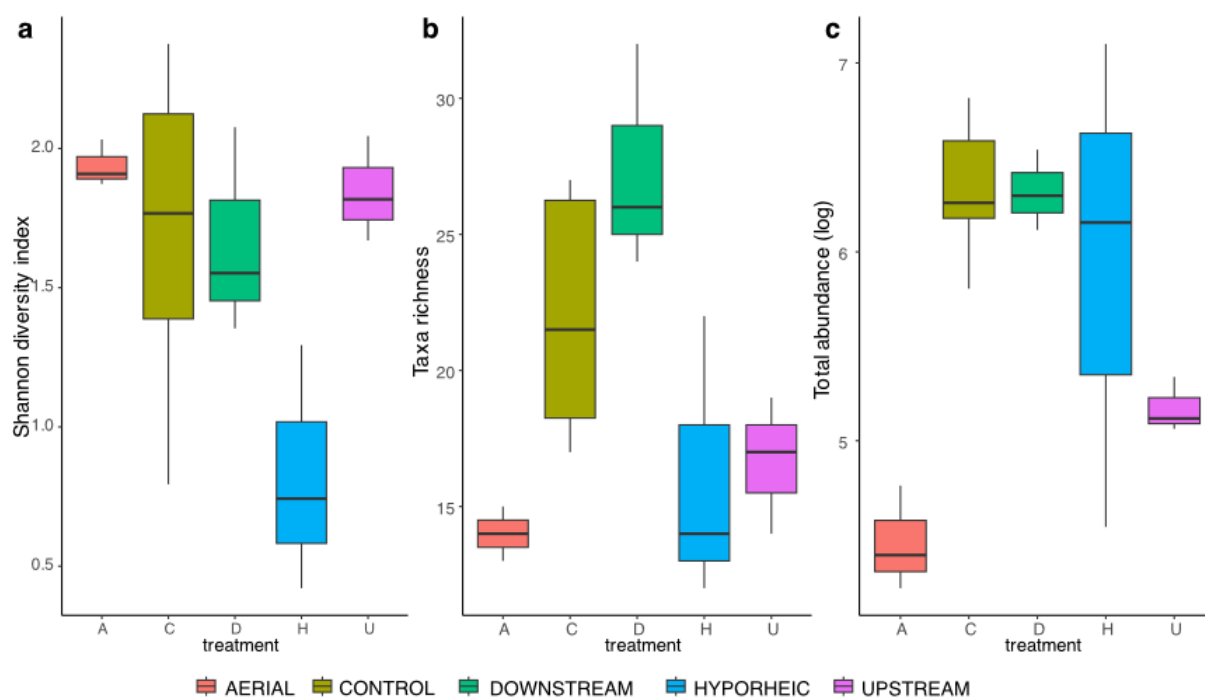


Fig 7. A comparison of Shannon diversity index (a), taxa richness (b) and total abundance (c) among dispersal treatments.

Total biomass was highest in the HYPORHEIC treatment (Fig 8a); however, it was composed of *Gammarus fossarum* biomass by almost 100 % (Fig 8b). In the other treatments, the contribution of *G. fossarum* species to the total biomass was lower.

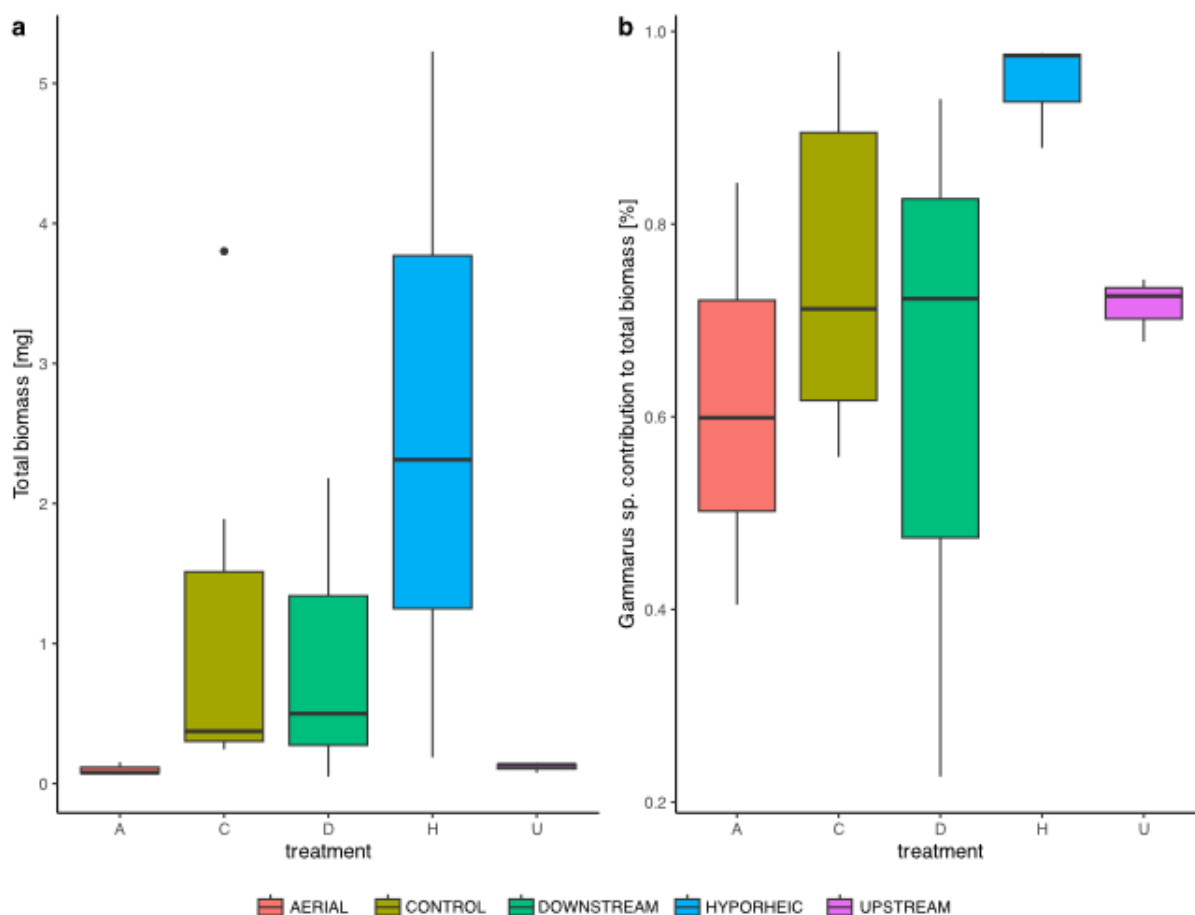


Fig 8. A comparison of total biomass in different dispersal treatments (a) and percentage of *Gammarus fossarum* contribution on the total biomass (b).

The analysis of species traits showed a different functional composition of the communities of each dispersal treatment. For example, HYPORHEIC treatments were more frequently colonised by species with lower body flexibility (Fig 9a), taxa with affinity to sand or gravel (psammal and akal; Fig 9b) or species typical for hyporheic zone (Fig 9c). AERIAL treatments were more frequently colonised by species with flexible life cycles (Fig 9d) and UPSTREAM treatments were more frequently colonised by species from lower stream zones or species typical for standing waters, whereas species from upper stream sections were less frequent (Fig 9e). All ANOVA results comparing the control treatment with the other treatments are shown in Table 1.

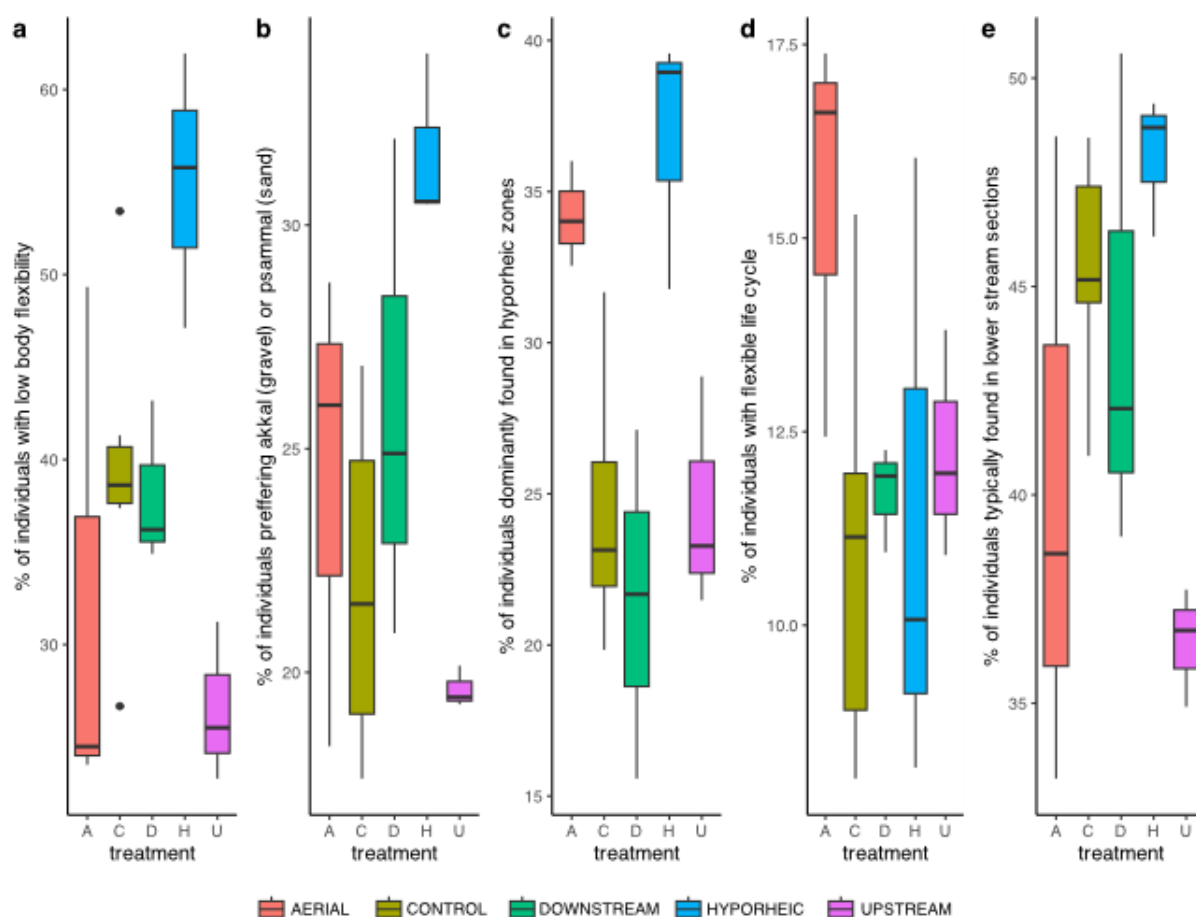


Fig 9. A comparison of selected invertebrate traits as functional representation of communities across dispersal treatments: low body flexibility (a), preference to gravel or sand (b), dominant presence in hyporheic zones (c), flexible life cycle (d) and preference to lower stream sections or stagnant waters (e).

Table 1. An overview of ANOVA results (estimate and p value) comparing the percentage representation of selected traits between control group and other treatments using defined contrasts. Non-significant results with p values larger than 0.05 (reported as n.s.) were greyed.

trait category	trait	control group vs. AERIAL		control group vs. DOWNSTREAM		control group vs. HYPORHEIC		control group vs. UPSTREAM	
		estimate	p value	estimate	p value	estimate	p value	estimate	p value
Body flexibility	none	2.10	n.s.	0.58	n.s.	1.31	n.s.	2.97	n.s.
	low	-6.89	n.s.	-1.23	n.s.	15.61	0.025	-12.84	n.s.
	high	4.79	n.s.	0.65	n.s.	-16.93	0.009	9.87	n.s.
Life cycle	semivoltine	-0.44	n.s.	2.78	0.039	-0.73	n.s.	-0.08	n.s.
	univoltine	-4.85	n.s.	0.61	n.s.	0.53	n.s.	-9.02	n.s.
	bivoltine	0.32	n.s.	-6.19	0.047	2.60	n.s.	6.14	0.049
	trivoltine	-1.10	n.s.	0.18	n.s.	-3.21	n.s.	1.00	n.s.
	multivoltine	1.58	n.s.	1.92	n.s.	0.39	n.s.	0.73	n.s.
	flexible	4.48	0.031	0.71	n.s.	0.42	n.s.	1.23	n.s.
Dissemination strategy	aquatic passive	3.97	n.s.	-0.07	n.s.	-3.67	n.s.	3.45	n.s.
	aquatic active	-3.15	n.s.	0.37	n.s.	7.96	0.021	-5.25	n.s.
	aerial passive	-0.24	n.s.	-3.34	n.s.	-2.65	n.s.	-0.51	n.s.
	aerial active	-0.58	n.s.	3.04	n.s.	-1.64	n.s.	2.32	n.s.
Occurrence in drift	catastrophic only	-3.40	n.s.	-2.58	n.s.	-1.18	n.s.	-0.68	n.s.
	typical	-7.59	n.s.	7.83	n.s.	-4.52	n.s.	-4.93	n.s.
	dominant	10.99	0.020	-5.24	n.s.	5.70	n.s.	5.61	n.s.
Microhabitat preference	pellal	-5.15	n.s.	2.56	n.s.	-4.30	n.s.	-1.90	n.s.
	psammal	1.73	n.s.	1.84	n.s.	3.55	0.024	-0.09	n.s.
	akkal	0.69	n.s.	2.13	n.s.	6.13	0.002	-2.21	n.s.
	particulated organic matter	1.69	n.s.	-2.33	n.s.	-2.64	n.s.	6.58	0.006
Hyporheal occurrence	accidental only	6.60	n.s.	-0.25	n.s.	-5.89	n.s.	3.24	n.s.
	common	-16.40	0.000	3.18	n.s.	-6.49	n.s.	-3.39	n.s.
	dominant	9.80	0.006	-2.93	n.s.	12.37	0.001	0.15	n.s.
Locomotion	swimming/diving	1.24	n.s.	-1.26	n.s.	5.25	0.004	0.93	n.s.
	burrowing	4.53	n.s.	5.43	0.032	1.58	n.s.	0.06	n.s.
	(semi-)sessil	-5.52	0.038	-0.01	n.s.	-4.85	n.s.	-2.28	n.s.
	other	0.17	n.s.	0.00	n.s.	0.07	n.s.	0.00	n.s.
Stream zonation	epirhithral	-2.47	0.033	-1.28	n.s.	0.63	n.s.	-3.91	0.002
	metarhithral	-2.81	n.s.	-0.24	n.s.	2.11	n.s.	-5.03	0.037
	hyporhithral	-0.64	n.s.	1.01	n.s.	1.29	n.s.	0.73	n.s.
	epipotamal	-0.05	n.s.	0.11	n.s.	-0.14	n.s.	1.05	0.019
	litoral	3.11	n.s.	-1.63	n.s.	-2.29	n.s.	4.96	0.017

Core experiment – destruction of the running experiment by the flash-flood

Based on the results of the pilot experiment, slight modifications were made to the flumes design to prevent, for example, gammarids intrusion into the aerial treatment. In mid-May, on the 25th day of the running experiment, an unexpected flash-flood (Fig 10) following a local extreme precipitation event (Fig 11) completely damaged nearly all boxes installed in both studied reaches of *Hrubý potok* and *Kazivec* streams (Fig 12). From 80 boxes installed in the stream at that moment, we only found a few, mostly damaged pieces (Fig 13). The low number of recovered functional boxes did not allow us to continue the experiment, as we spent all allocated budget to the construction of expensive stainless-steel boxes, plastic and rubber sealing, tailored netting, gravel material, and dataloggers. Additionally, repeated manufacturing of the hand-crafted equipment would have taken many months at much higher than initial costs due to rapidly increasing market prices of material. An experiment repetition within the context of DRYVER was therefore deemed impossible. Both model reaches were

subsequently inspected several times to remove all destroyed remains of boxes from the streams, to avoid any negative effects on streams in the protected area of White Carpathians.

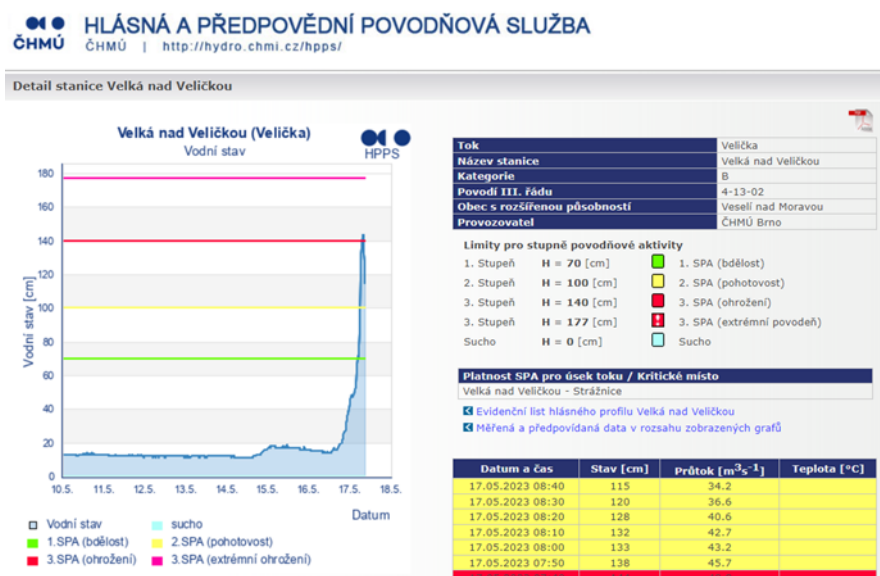


Fig 10. Rise of large flood (50-year frequency) on the May 17, 2023 when water depth increases 7 times. (Print screen from Czech hydro-meteorological institute; chmi.cz)

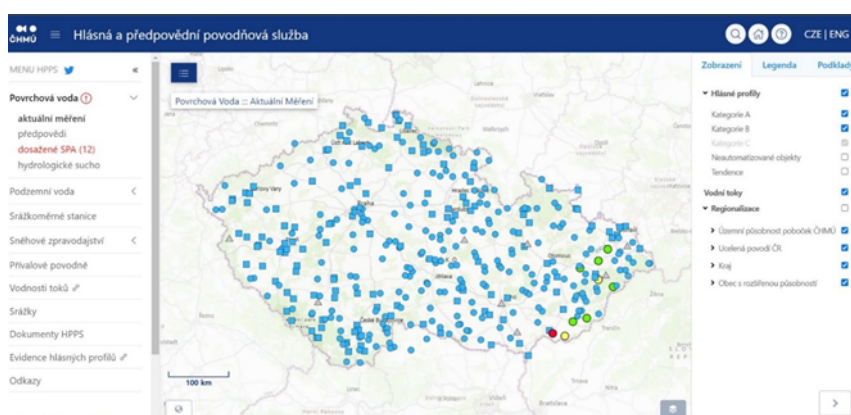


Fig 11. The only red dot (highest flood event) on map is the Velička catchment. The rest of the Czech Republic shows “normal” discharge. Screenshot from May 17, 2023. (Print screen from Czech hydro-meteorological institute; chmi.cz)



Fig 12. Perennial stream section with installed boxes BEFORE (left) and DURING (right) the flash flood event May 17, 2023. Pictures taken by installed phototraps monitoring the experiment several times a day.



Fig 13. Remains of damaged boxes accumulated in the riparian vegetation (left) and squeezed unusable box (right).

Conclusions

Even though the experiment was destroyed by a flood, the functionality of the proposed solution and the ability of the developed flumes designed to capture different recolonization pathways was demonstrated in a preceding short pilot experiment. While drift was the most favourable colonisation pathway (compared to the control group) in terms of biodiversity, the hyporheic pathway was the most important recolonization pathway in terms of biomass. The aerial recolonization route played only a marginal role, however, due to the short duration of the experiment, species that would have hatched from eggs laid by flying adults could not develop sufficiently. Recolonization from downstream (i.e. UPSTREAM treatment) is likely to play only a marginal role in stream recovery both in terms of diversity and biomass. However, the entire pilot experiment took place in a perennial stream only, and it is to be expected that when evaluating an experiment from a non-perennial stream, where communities are already largely affected by drying, the importance of different recolonization routes would vary. Despite the main experiment collapsing during floods, the promising results of the pilot study ensure that the proposed experimental design should serve for potential future experiments to evaluate the effects of flow intermittence to regulate functional recovery of aquatic metacommunities and in-stream ecosystem functioning.

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